

# Term 3 Practice Activity: C1 Density Functions

## Contents

- 1 (5) First-Investment Product Resale Plans (C1)
- 2 (11) Online Coffee Sales Investment (C1)
- 3 (1) Checking Three Candidate Density Functions I (C1)
- 4 (2) Checking Two Candidate Density Functions II (C1)
- 5 (4) Checking Two Candidate Density Functions III (C1)
- 6 (2) Battery Charging Time (C1)
- 7 (3) Startup Delay (C1)
- 8 (5) Packing Score (C1)
- 9 (12) Two-Scenario Inventory Investment (C2)
- 10 (3) Uniform Distribution I (C2)
- 11 (5) Uniform Distribution II (C2)
- 12 (4) Two-Interval Piecewise Uniform (C2)
- 13 (5) Two-Interval Piecewise Uniform with Unique Variable Endpoint (C2)
- 14 (4) Piecewise Uniform with Unknown  $k$  (C2)
- 15 (12) Small-Business Investment (C3)
- 16 (11) Basic Symmetric Triangular Density (C3)
- 17 (12) Introductory Linear Density (C3)

## Evaluated criteria

- C1. Describes probabilities in continuous variables as a density function.
- C2. Explains how a uniform random variable works..
- C3. Expresses how to work with distributions that have linear behaviour.

## 1 (5) First-Investment Product Resale Plans (C1)

An entrepreneur is comparing four product resale plans. The required initial investments are 10, 20, 30, and 40 million pesos. For each plan, the end-of-quarter value is modeled as uniformly distributed on a projected interval.

For the first plan, the final value is between 8 and 16 million pesos. For the second plan, it is between 16 and 28 million pesos. For the third plan, it is between 24 and 36 million pesos. For the fourth plan, it is between 30 and 45 million pesos.

A *loss* occurs when the final value is below the amount invested, and an *earning* occurs when the final value is above it.

### Questions/Tasks.

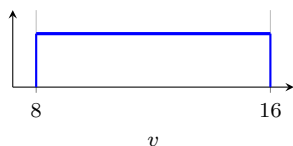
- a) (2) Write the density constant for each investment plan.
- b) (3) Find the probability of loss and the probability of earning for each investment plan. (0.5) Then identify which investment has the highest probability of generating losses and which investment has the highest probability of generating earnings.

### a) (2) Write the density constant for each investment plan.

For a uniform distribution on  $[a, b]$ , the density is

$$f(v) = \frac{1}{b-a} \quad \text{for } a \leq v \leq b.$$

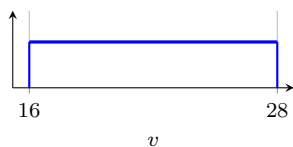
Therefore,



**First investment option.**

$$f_{V_1}(v) = \frac{1}{16-8} \quad \longrightarrow \quad f_{V_1}(v) = \frac{1}{8}$$

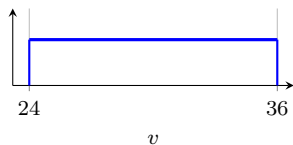
$$f_{V_1}(v) = \frac{1}{8} \quad \text{for } 8 \leq v \leq 16.$$



**Second investment option.**

$$f_{V_2}(v) = \frac{1}{28-16} \quad \longrightarrow \quad f_{V_2}(v) = \frac{1}{12}$$

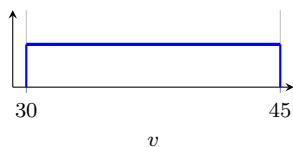
$$f_{V_2}(v) = \frac{1}{12} \quad \text{for } 16 \leq v \leq 28.$$



**Third investment option.**

$$f_{V_3}(v) = \frac{1}{36-24} \quad \longrightarrow \quad f_{V_3}(v) = \frac{1}{12}$$

$$f_{V_3}(v) = \frac{1}{12} \quad \text{for } 24 \leq v \leq 36.$$



**Fourth investment option.**

$$f_{V_4}(v) = \frac{1}{45-30} \quad \longrightarrow \quad f_{V_4}(v) = \frac{1}{15}$$

$$f_{V_4}(v) = \frac{1}{15} \quad \text{for } 30 \leq v \leq 45.$$

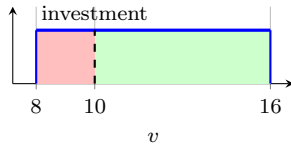


**b) (3) Find the probability of loss and the probability of earning for each investment plan. (0.5) Then identify which investment has the highest probability of generating losses and which investment has the highest probability of generating earnings.**

Since these are continuous models, the probability of ending with exactly the invested amount is 0. So we compare interval lengths.

**First investment option.**

Loss



$$P(V_1 < 10) = \frac{10 - 8}{16 - 8} \quad \longrightarrow \quad P(V_1 < 10) = \frac{2}{8}$$

$$\longrightarrow \quad P(V_1 < 10) = \frac{1}{4}.$$

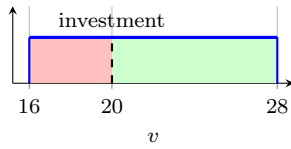
Earning

$$P(V_1 > 10) = \frac{16 - 10}{16 - 8} \quad \longrightarrow \quad P(V_1 > 10) = \frac{6}{8}$$

$$\longrightarrow \quad P(V_1 > 10) = \frac{3}{4}.$$

**Second investment option.**

Loss



$$P(V_2 < 20) = \frac{20 - 16}{28 - 16} \quad \longrightarrow \quad P(V_2 < 20) = \frac{4}{12}$$

$$\longrightarrow \quad P(V_2 < 20) = \frac{1}{3}.$$

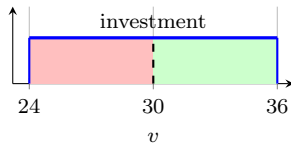
Earning

$$P(V_2 > 20) = \frac{28 - 20}{28 - 16} \quad \longrightarrow \quad P(V_2 > 20) = \frac{8}{12}$$

$$\longrightarrow \quad P(V_2 > 20) = \frac{2}{3}.$$

**Third investment option.**

Loss



$$P(V_3 < 30) = \frac{30 - 24}{36 - 24} \quad \longrightarrow \quad P(V_3 < 30) = \frac{6}{12}$$

$$\longrightarrow \quad P(V_3 < 30) = \frac{1}{2}.$$

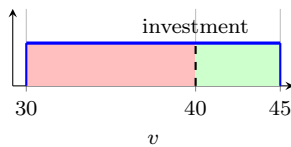
Earning

$$P(V_3 > 30) = \frac{36 - 30}{36 - 24} \quad \longrightarrow \quad P(V_3 > 30) = \frac{6}{12}$$

$$\longrightarrow \quad P(V_3 > 30) = \frac{1}{2}.$$

**Fourth investment option.**

Loss



$$P(V_4 < 40) = \frac{40 - 30}{45 - 30} \quad \longrightarrow \quad P(V_4 < 40) = \frac{10}{15}$$

$$\longrightarrow \quad P(V_4 < 40) = \frac{2}{3}.$$

Earning

$$P(V_4 > 40) = \frac{45 - 40}{45 - 30} \quad \longrightarrow \quad P(V_4 > 40) = \frac{5}{15}$$

$$\longrightarrow \quad P(V_4 > 40) = \frac{1}{3}.$$

Comparing the loss probabilities

$$\frac{1}{4}, \quad \frac{1}{3}, \quad \frac{1}{2}, \quad \frac{2}{3},$$

the largest is

$$\frac{2}{3}.$$

So the investment with the highest probability of generating losses is the fourth investment option, which requires 40 million pesos.

Comparing the earning probabilities

$$\frac{3}{4}, \quad \frac{2}{3}, \quad \frac{1}{2}, \quad \frac{1}{3},$$

the largest is

$$\frac{3}{4}.$$

So the investment with the highest probability of generating earnings is the first investment option, which requires 10 million pesos.



## 2 (11) Online Coffee Sales Investment (C1)

A small entrepreneur runs a specialty coffee resale business. Let  $V$  be the end-of-quarter value, measured in million pesos. The final value cannot be below 30 and cannot exceed 60. Lower values are possible, but higher values become gradually more likely across this interval, so the final value is modeled with an increasing density on  $[30, 60]$ . The initial investment is 40 million pesos. The probability of the extreme value of 30 million pesos can be considered to be 0.

**Questions/Tasks.**

- a) (3) Determine the density constant and write the pdf.
- b) (2) Find the probability of loss and the probability of earning.
- c) (4) Compute the probability that the money obtained falls between 40 and 50 million.
- d) (2) Compute the probability that the money obtained is above 50 million.

**a) (3) Determine the density constant and write the pdf.**

Let  $V$  denote the end-of-quarter value, measured in million pesos. Since the density rises linearly from the lower endpoint 30 to the upper endpoint 60, we model it by

$$f(v) = k(v - 30) \quad \text{for } 30 \leq v \leq 60,$$

and

$$f(v) = 0 \quad \text{otherwise.}$$

To determine  $k$ , use total area = 1. The graph is a straight line from  $(30, 0)$  up to  $(60, 30k)$ , so the region under the density is a triangle with base 30 and height  $30k$ . Therefore,

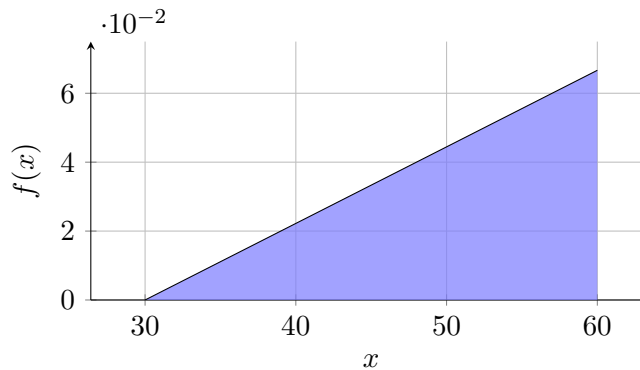
$$\frac{1}{2}(30)(30k) = 1.$$

So

$$\frac{1}{2}(30)(30k) = 1 \quad \longrightarrow \quad 450k = 1 \quad \longrightarrow \quad k = \frac{1}{450}.$$

Hence the pdf is

$$f(v) = \frac{1}{450}(v - 30) \quad \text{for } 30 \leq v \leq 60.$$



b) (2) Find the probability of loss and the probability of earning.

**Probability of loss.** A loss occurs when the final value is below the purchase price of 40 million pesos. On the interval from 30 to 40, the graph forms a smaller triangle.

Its base is

$$40 - 30 = 10,$$

and its height is

$$f(40) = \frac{1}{450}(40 - 30) = \frac{10}{450} = \frac{1}{45}.$$

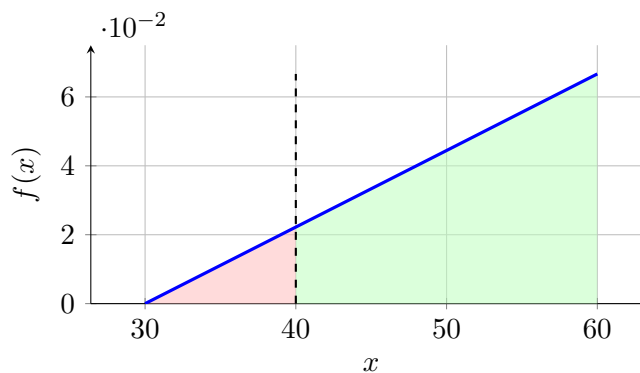
So

$$P(\text{loss}) = P(V < 40) = \frac{1}{2}(10) \left( \frac{1}{45} \right) \quad \longrightarrow \quad P(\text{loss}) = \frac{10}{90} \quad \longrightarrow \quad P(\text{loss}) = \frac{1}{9}.$$

**Probability of earning.** An earning occurs when the final value is above the purchase price of 40 million pesos. Since the total probability is 1 and a loss occurs when the final value is below 40, we use the complement rule

$$P(\text{earning}) = P(V > 40) = 1 - P(V < 40).$$

$$\text{Loss} \rightarrow P(V < 40) = \frac{1}{9} \quad \text{Earning} \rightarrow P(V > 40) = 1 - \frac{1}{9} = \frac{8}{9}$$





c) (4) Compute the probability that the money obtained falls between 40 and 50 million.

Now compute the probability that the money obtained falls between 40 and 50 million pesos.

From part b),

$$P(V > 40) = \frac{8}{9}.$$

From part d),

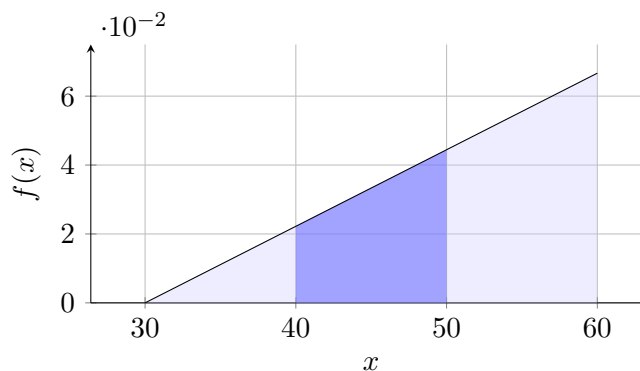
$$P(V > 50) = \frac{5}{9}.$$

Therefore, using the difference of these two events,

$$\begin{aligned} P(40 \leq V \leq 50) &= P(V > 40) - P(V > 50) &\longrightarrow & P(40 \leq V \leq 50) = \frac{8}{9} - \frac{5}{9} \\ &&\longrightarrow & P(40 \leq V \leq 50) = \frac{3}{9} &\longrightarrow & P(40 \leq V \leq 50) = \frac{1}{3}. \end{aligned}$$

Thus, the probability that the money obtained falls between 40 and 50 million pesos is

$$\frac{1}{3}.$$



d) (2) Compute the probability that the money obtained is above 50 million.

Now compute the probability that the money obtained is above 50 million pesos.

Instead of calculating the small region to the right of 50, it is easier to first calculate the probability that the money obtained is less than 50 million pesos. On the interval from 30 to 50, the graph forms a triangle.

Its base is

$$50 - 30 = 20,$$

and its height is

$$f(50) = \frac{1}{450}(50 - 30) = \frac{20}{450} = \frac{2}{45}.$$

So

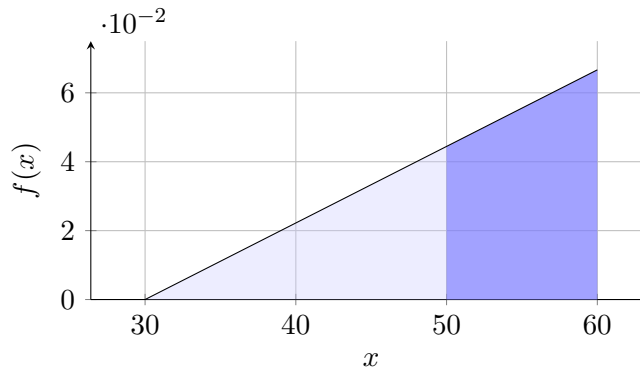
$$P(V < 50) = \frac{1}{2}(20) \left( \frac{2}{45} \right) \longrightarrow P(V < 50) = \frac{20}{45} \longrightarrow P(V < 50) = \frac{4}{9}.$$

Using the complement rule,

$$P(V > 50) = 1 - P(V < 50) \longrightarrow P(V > 50) = 1 - \frac{4}{9} \longrightarrow P(V > 50) = \frac{5}{9}.$$

Thus, the probability that the money obtained is above 50 million pesos is

$$\frac{5}{9}.$$



### 3 (1) Checking Three Candidate Density Functions I (C1)

A continuous random variable is modeled on the interval shown in each case.

$$\text{Distribution A: } f(x) = 0.20 \quad \text{for } 0 \leq x \leq 5$$

$$\text{Distribution B: } f(x) = 0.30 \quad \text{for } 0 \leq x \leq 5$$

$$\text{Distribution C: } f(x) = -0.20 \quad \text{for } 0 \leq x \leq 5$$

**Questions/Tasks.**

a) (0.5) Decide which distribution is a valid PDF and which ones are not. Justify your answer.

b) (0.5) For the valid distribution, find  $P(1 \leq X \leq 4)$ .

a) (0.5) Decide which distribution is a valid PDF and which ones are not. Justify your answer.

For a valid PDF,  $f(x) \geq 0$  on its support and total area is 1.

- Distribution A: support  $[0, 5]$ , constant height 0.20.

$$\text{Area} = 5(0.20) = 1,$$

so A is valid.



- Distribution B: support  $[0, 5]$ , constant height 0.30.

$$\text{Area} = 5(0.30) = 1.5 \neq 1,$$

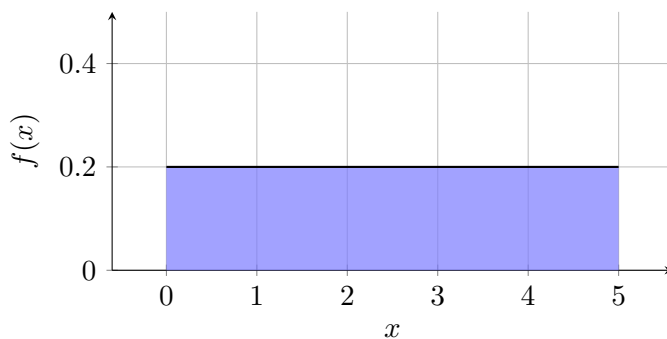
so B is not valid.

- Distribution C: support  $[0, 5]$ , constant height  $-0.20$ . Since

$$-0.20 < 0,$$

this function is negative on its support, so C is not valid.

### interpretation



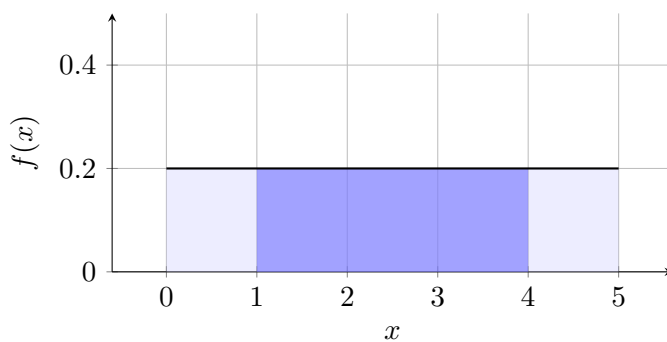
The valid PDF is uniform on  $[0, 5]$ : it is constant, nonnegative, and gives equal probability to equal-width intervals.



**b) (0.5) For the valid distribution, find  $P(1 \leq X \leq 4)$ .**

For the valid distribution A,

$$P(1 \leq X \leq 4) = (4 - 1)(0.20) \quad \longrightarrow \quad P(1 \leq X \leq 4) = 3(0.20) \quad \longrightarrow \quad P(1 \leq X \leq 4) = 0.6.$$



## 4 (2) Checking Two Candidate Density Functions II (C1)

A continuous random variable is modeled on the interval shown in each case.

$$\text{Distribution A} \quad f(x) = \frac{x}{18} \quad \text{for } 0 \leq x \leq 6$$

$$\text{Distribution A} \quad f(x) = \frac{x}{12} \quad \text{for } 0 \leq x \leq 6$$

### Questions/Tasks.

a) (1) Decide which distribution is a valid PDF and which one is not. Justify your answer.

b) (1) For the valid distribution, find  $P(3 \leq X \leq 6)$ .

a) (1) Decide which distribution is a valid PDF and which one is not. Justify your answer.

Check nonnegativity and total area on support  $[0, 6]$ .

- Distribution A:  $f(x) = \frac{x}{18}$  forms a triangle with base 6 and height  $\frac{6}{18} = \frac{1}{3}$ .

$$\text{Area} = \frac{1}{2}(6) \left( \frac{1}{3} \right) = 1,$$

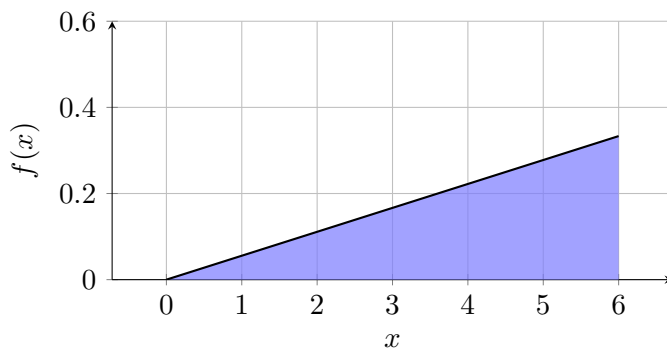
so A is valid.

- Distribution B:  $f(x) = \frac{x}{12}$  has base 6 and height  $\frac{6}{12} = \frac{1}{2}$ .

$$\text{Area} = \frac{1}{2}(6) \left( \frac{1}{2} \right) = \frac{3}{2} \neq 1,$$

so B is not valid.

### Interpretation



The valid PDF increases linearly on  $[0, 6]$ , so larger  $x$  values have higher density than smaller ones.



b) (1) For the valid distribution, find  $P(3 \leq X \leq 6)$ .

For A,

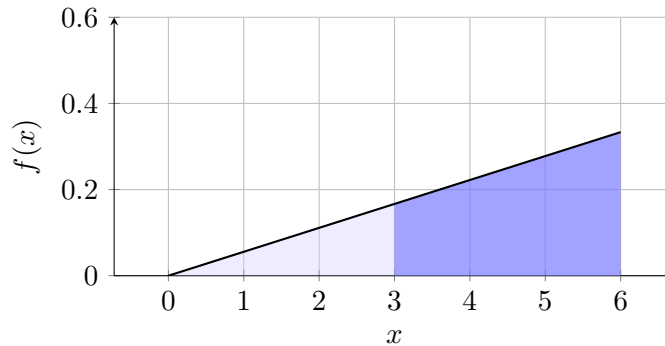
$$P(3 \leq X \leq 6) = 1 - P(0 \leq X \leq 3).$$

On  $[0, 3]$ , area is a triangle with base 3 and height  $\frac{3}{18} = \frac{1}{6}$ :

$$P(0 \leq X \leq 3) = \frac{1}{2}(3) \left( \frac{1}{6} \right) \longrightarrow P(0 \leq X \leq 3) = \frac{3}{12} \longrightarrow P(0 \leq X \leq 3) = \frac{1}{4}.$$

Hence

$$P(3 \leq X \leq 6) = 1 - \frac{1}{4} \longrightarrow P(3 \leq X \leq 6) = \frac{3}{4}.$$



## 5 (4) Checking Two Candidate Density Functions III (C1)

A continuous random variable is modeled on the interval shown in each case.

$$\text{Distribution A} \quad f(x) = \frac{6-x}{12} \quad \text{for } 1 \leq x \leq 5$$

$$\text{Distribution B} \quad f(x) = \frac{6-x}{10} \quad \text{for } 1 \leq x \leq 5$$

Questions/Tasks.

a) (2) Decide which distribution is a valid PDF and which one is not.

b) (2) For the valid distribution, find  $P(1 \leq X \leq 3)$ .

a) (2) Decide which distribution is a valid PDF and which one is not.

On support  $[1, 5]$ , both densities are nonnegative. Check total area:

- Distribution A:  $f(x) = \frac{6-x}{12}$  gives a trapezium with base 4, left height  $f(1) = \frac{5}{12}$ , and right height  $f(5) = \frac{1}{12}$ .

$$\text{Area} = \frac{\frac{5}{12} + \frac{1}{12}}{2} \cdot 4 = 1,$$

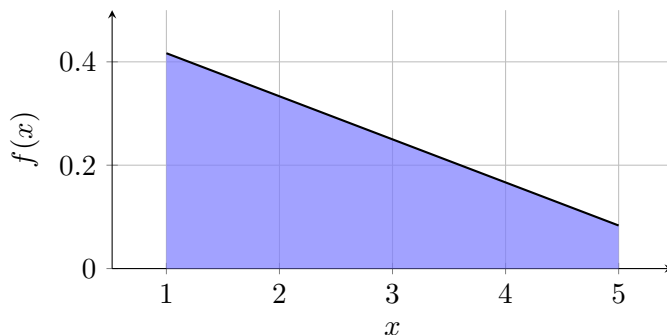
so A is valid.

- Distribution B:  $f(x) = \frac{6-x}{10}$  gives base 4, left height  $f(1) = \frac{5}{10} = \frac{1}{2}$ , and right height  $f(5) = \frac{1}{10}$ .

$$\text{Area} = \frac{\frac{1}{2} + \frac{1}{10}}{2} \cdot 4 = \frac{6}{5} \neq 1,$$

so B is not valid.

### Interpretation



The valid PDF is decreasing on  $[1, 5]$ , so values near 1 are more likely than values near 5, but values near 5 still have positive density.



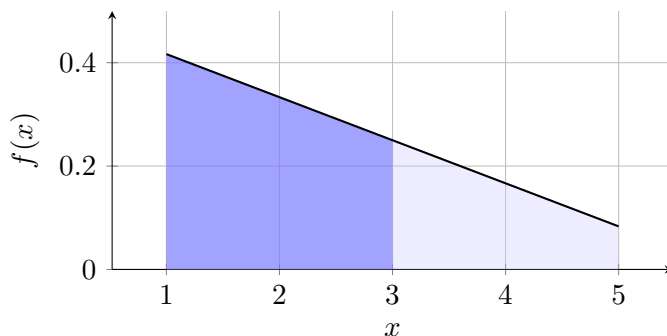
b) (2) For the valid distribution, find  $P(1 \leq X \leq 3)$ .

For valid distribution A,

$$P(1 \leq X \leq 3) = \text{area of trapezium on } [1, 3].$$

With heights  $f(1) = \frac{5}{12}$  and  $f(3) = \frac{1}{4}$ , width 2:

$$P(1 \leq X \leq 3) = \frac{\frac{5}{12} + \frac{1}{4}}{2} \cdot 2 = \frac{2}{3}.$$



**Final answer:** A is valid, B is not valid, and  $P(1 \leq X \leq 3) = \frac{2}{3}$ .



## 6 (2) Battery Charging Time (C1)

Let  $X$  be the battery charging time (in h) with support  $[1, 5]$  and density

$$f(x) = k \quad \text{for } 1 \leq x \leq 5.$$

**Questions/Tasks.**

a) (1) Find  $k$  so  $f(x)$  is a valid PDF.

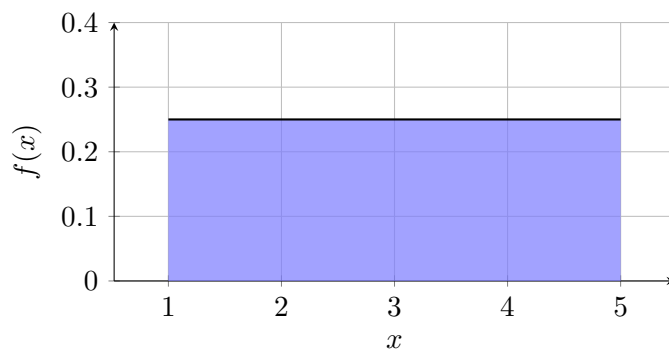
b) (1) Find  $P(2 \leq X \leq 4)$ .

a) (1) Find  $k$  so  $f(x)$  is a valid PDF.

Since  $f(x) = k$  on  $[1, 5]$ , total area is rectangle area:

$$4k = 1 \quad \longrightarrow \quad k = \frac{1}{4}.$$

**Interpretation**



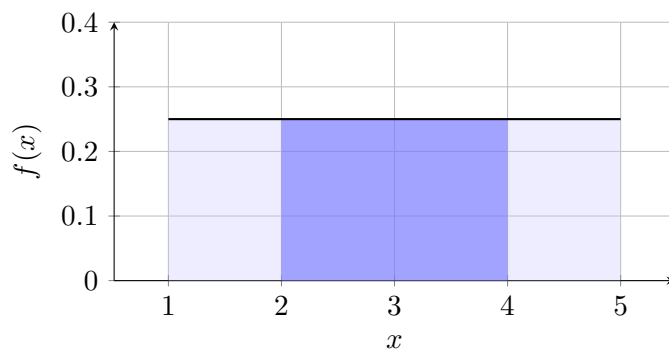
The PDF graph shows the support and overall shape of the density on that support.



b) (1) Find  $P(2 \leq X \leq 4)$ .

For the requested probability,

$$P(2 \leq X \leq 4) = \text{width} \times \text{height} = (4-2) \left( \frac{1}{4} \right) \quad \longrightarrow \quad P(2 \leq X \leq 4) = 2 \left( \frac{1}{4} \right) \quad \longrightarrow \quad P(2 \leq X \leq 4) = \frac{1}{2}.$$





## 7 (3) Startup Delay (C1)

Let  $T$  be a startup delay (in s) with support  $[0, 8]$  and density

$$f(t) = kt \quad \text{for } 0 \leq t \leq 8.$$

**Questions/Tasks.**

a) (1) Find  $k$  so  $f(t)$  is a valid PDF.

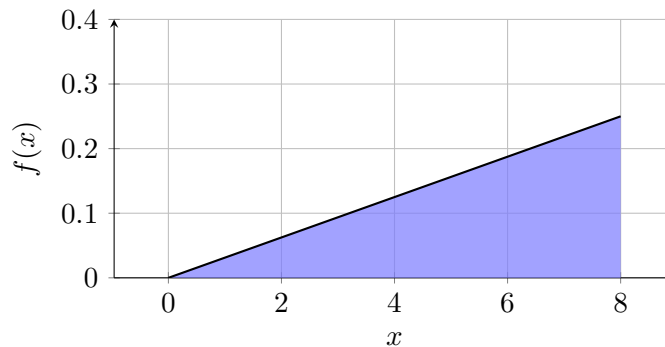
b) (2) Find  $P(T \geq 6)$ .

a) (1) Find  $k$  so  $f(t)$  is a valid PDF.

For  $f(t) = kt$  on  $[0, 8]$ , area is a triangle with base 8 and height  $8k$ :

$$\frac{1}{2}(8)(8k) = 32k \quad \longrightarrow \quad 32k = 1 \quad \longrightarrow \quad k = \frac{1}{32}.$$

**Interpretation**



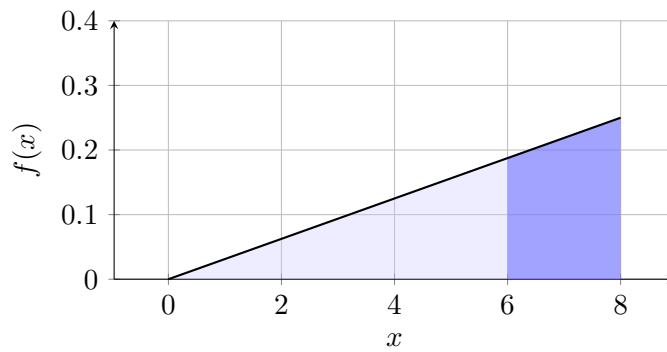
The PDF graph shows the support and overall shape of the density on that support.



b) (2) Find  $P(T \geq 6)$ .

For  $P(T \geq 6)$ , use area from 6 to 8 (a trapezium): heights are  $6k$  and  $8k$ , width 2.

$$P(T \geq 6) = \frac{6k + 8k}{2} \cdot 2 \quad \longrightarrow \quad P(T \geq 6) = 14k \quad \longrightarrow \quad P(T \geq 6) = \frac{14}{32} \quad \longrightarrow \quad P(T \geq 6) = \frac{7}{16}.$$



## 8 (5) Packing Score (C1)

**Problem.** Let  $Q$  be a packing score with support  $[2, 6]$  and density

$$f(q) = k(q - 1) \quad \text{for } 2 \leq q \leq 6.$$

**Questions/Tasks.**

a) (3) Find  $k$  so  $f(q)$  is a valid PDF.

b) (2) Find  $P(3 \leq Q \leq 5)$ .

a) (3) Find  $k$  so  $f(q)$  is a valid PDF.

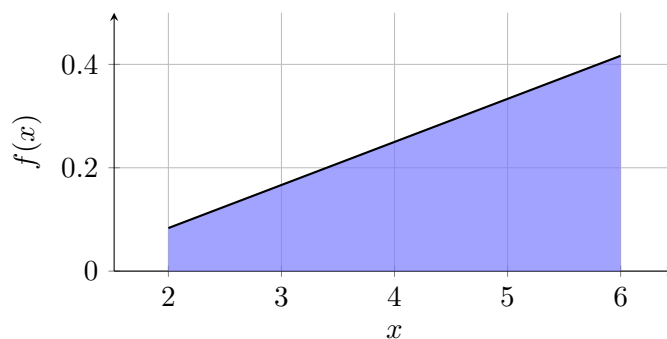
On  $[2, 6]$ ,  $f(q) = k(q - 1)$  is linear and positive. Total area is a trapezium with width 4, heights

$$f(2) = k, \quad f(6) = 5k.$$

So

$$\frac{k + 5k}{2} \cdot 4 = 12k \quad \longrightarrow \quad 12k = 1 \quad \longrightarrow \quad k = \frac{1}{12}.$$

**Interpretation**



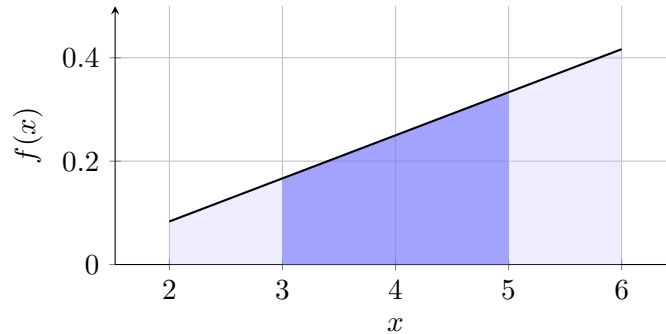
b) (2) Find  $P(3 \leq Q \leq 5)$ .

For

$$P(3 \leq Q \leq 5) = \text{trapezium area on } [3, 5],$$

with heights  $f(3) = 2k$ ,  $f(5) = 4k$ , width 2:

$$P(3 \leq Q \leq 5) = \frac{2k + 4k}{2} \cdot 2 \quad \longrightarrow \quad P(3 \leq Q \leq 5) = 6k \quad \longrightarrow \quad P(3 \leq Q \leq 5) = \frac{1}{2}.$$



**Final answer:**  $k = \frac{1}{12}$  and  $P(3 \leq Q \leq 5) = \frac{1}{2}$ .



## 9 (12) Two-Scenario Inventory Investment (C2)

**Problem description:** A seller invests in a short-term inventory deal to resell packaged foods in nearby cities. For the quarter, they consider two realistic market scenarios. In Scenario 1, transport bottlenecks continue and restocking stays slow; this has probability 0.4. In Scenario 2, transport returns to normal and inventory rotates faster; this has probability 0.6. If Scenario 1 occurs, the final value of the investment is expected to end between fifty and seventy million pesos, with all values in that interval treated as equally plausible. If Scenario 2 occurs, the final value is expected to end between sixty and ninety million pesos, again with all values in that interval treated as equally plausible. The initial investment is sixty-five million pesos. A loss occurs if the final value is below that amount, and an earning occurs if the final value is above it.

**Questions/Tasks.**

- (2) Write the conditional density for each scenario.
- (3) Build the unconditional pdf of the final value as a piecewise function.
- (2) Verify that the unconditional pdf integrates to one.
- (3) Find the probability of loss and the probability of earning.
- (2) Find the probability that the final value falls inside the overlap interval.

A conditional density function describes how the final value is distributed once one specific scenario is assumed to be true. In this problem, each scenario has its own density because the investment behaves differently depending on



whether transport problems continue or normal logistics return. So, the conditional density tells us how the final value is distributed under that particular market condition.

The general, or unconditional, density function combines those scenario-specific densities into one overall model. It describes the distribution of the final value before knowing which scenario actually occurs, and it is obtained by weighting the conditional densities by the probabilities of the scenarios. The overlap interval is especially important because it contains values that are plausible in both market scenarios. For a small investor, this shared middle range is often more useful for planning than extreme values that appear only under one scenario, since it represents outcomes that remain feasible whether logistics stay difficult or improve.

**a) (2) Write the conditional density for each scenario.**

Let  $V$  be the final value of the investment, measured in million pesos.

Under the delayed-clearance scenario, the value is uniformly distributed on  $[50, 70]$ . Therefore, the conditional density is

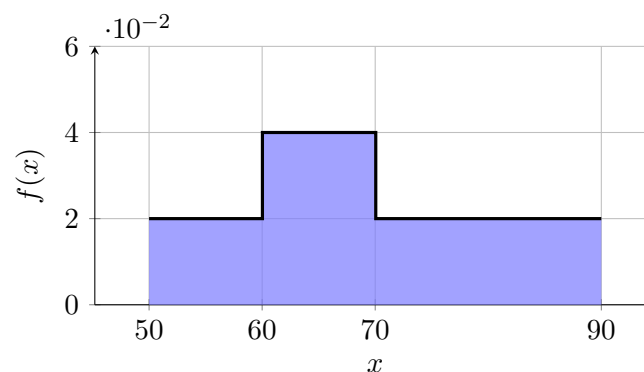
$$f_{V|S_1}(v) = \begin{cases} \frac{1}{20}, & 50 \leq v \leq 70, \\ 0, & \text{otherwise.} \end{cases}$$

Under the normal-clearance scenario, the value is uniformly distributed on  $[60, 90]$ . Therefore, the conditional density is

$$f_{V|S_2}(v) = \begin{cases} \frac{1}{30}, & 60 \leq v \leq 90, \\ 0, & \text{otherwise.} \end{cases}$$

The scenario probabilities are

$$P(S_1) = 0.4 \quad \text{and} \quad P(S_2) = 0.6.$$



So the unconditional density is the mixture

$$f_V(v) = 0.4 f_{V|S_1}(v) + 0.6 f_{V|S_2}(v).$$



**b) (3) Build the unconditional pdf of the final value as a piecewise function.**

Now build it by intervals.

For  $50 \leq v < 60$ , only Scenario 1 contributes, so

$$f_V(v) = 0.4 \left( \frac{1}{20} \right) = \frac{1}{50}.$$

For  $60 \leq v \leq 70$ , both scenarios contribute, so

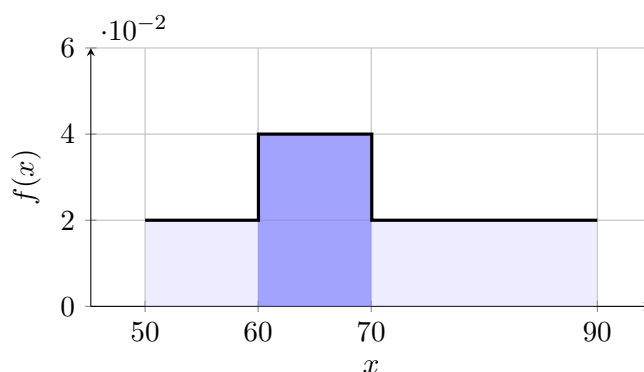
$$f_V(v) = 0.4 \left( \frac{1}{20} \right) + 0.6 \left( \frac{1}{30} \right) = \frac{1}{50} + \frac{1}{50} = \frac{1}{25}.$$

For  $70 < v \leq 90$ , only Scenario 2 contributes, so

$$f_V(v) = 0.6 \left( \frac{1}{30} \right) = \frac{1}{50}.$$

Therefore, the unconditional pdf is

$$f_V(v) = \begin{cases} \frac{1}{50}, & 50 \leq v < 60, \\ \frac{1}{25}, & 60 \leq v \leq 70, \\ \frac{1}{50}, & 70 < v \leq 90, \\ 0, & \text{otherwise.} \end{cases}$$



**c) (2) Verify that the unconditional pdf integrates to one.**

Use the total area of the three rectangles.

From 50 to 60, the width is 10 and the height is  $\frac{1}{50}$ , so the area is

$$10 \left( \frac{1}{50} \right) = \frac{1}{5}.$$

From 60 to 70, the width is 10 and the height is  $\frac{1}{25}$ , so the area is

$$10 \left( \frac{1}{25} \right) = \frac{2}{5}.$$

From 70 to 90, the width is 20 and the height is  $\frac{1}{50}$ , so the area is

$$20 \left( \frac{1}{50} \right) = \frac{2}{5}.$$

Adding the three pieces,

$$\frac{1}{5} + \frac{2}{5} + \frac{2}{5} = 1.$$

So the unconditional density is valid.



**d) (3) Find the probability of loss and the probability of earning.**

**Probability of loss.** A loss occurs when the final value is below the purchase price of 65 million pesos. This includes the whole interval from 50 to 60, plus the interval from 60 to 65.

From 50 to 60, the area is

$$10 \left( \frac{1}{50} \right) = \frac{1}{5}.$$

From 60 to 65, the width is 5 and the height is  $\frac{1}{25}$ , so the area is

$$5 \left( \frac{1}{25} \right) = \frac{1}{5}.$$

Therefore,

$$P(\text{loss}) = P(V < 65) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}.$$

**Probability of earning.** An earning occurs when the final value is above the purchase price of 65 million pesos. Since the total probability is 1 and a loss occurs when the final value is below 65, we use the complement rule

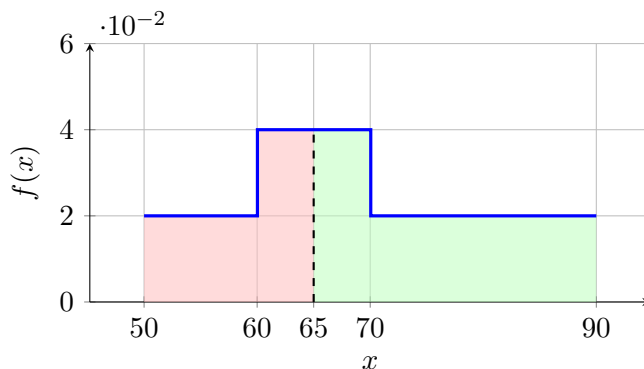
$$P(\text{earning}) = P(V > 65) = 1 - P(V < 65).$$

From the previous result,

$$P(V < 65) = \frac{2}{5}.$$

Therefore,

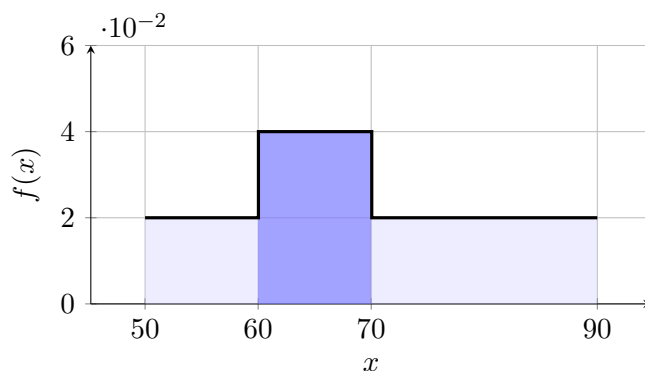
$$P(V > 65) = 1 - \frac{2}{5} \quad \longrightarrow \quad P(V > 65) = \frac{3}{5}.$$



e) (2) Find the probability that the final value falls inside the overlap interval.

The overlap interval is from 60 to 70. Over this region, the density is constant at  $\frac{1}{25}$ , so

$$P(60 \leq V \leq 70) = 10 \left( \frac{1}{25} \right) = \frac{2}{5}.$$



## 10 (3) Uniform Distribution I (C2)

Let  $X \sim \text{Uniform}(1, b)$ , where  $b > 8$ .

**Questions/Tasks.**

a) (1) Find the density value  $f(x)$  on the support,

b) (1) Find  $P(2 \leq X \leq 7)$ .

c) (1) Find  $P(2 \leq X \leq 7)$  when  $b = 9$  and  $b = 16$

a) (1) Find the density value  $f(x)$  on the support,

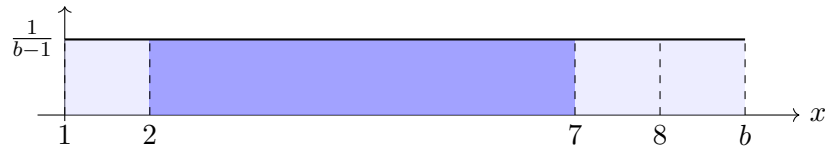
For  $X \sim \text{Uniform}(1, b)$ ,

$$f(x) = \frac{1}{b-1}, \quad 1 \leq x \leq b.$$



b) (1) Find  $P(2 \leq X \leq 7)$ .

$$P(2 \leq X \leq 7) = \frac{7-2}{b-1} = \frac{5}{b-1}.$$

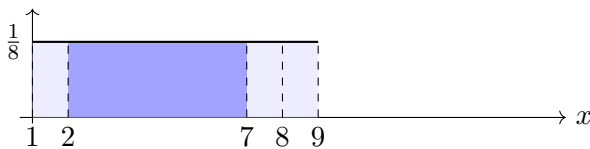


c) (1) Find  $P(2 \leq X \leq 7)$  when  $b = 9$  and  $b = 16$

Consider the following cases obtained by assigning specific values to  $b$ .

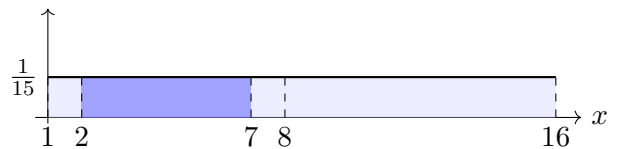
*Example with  $b = 9$*

$$P(2 \leq X \leq 7) = \frac{5}{8}$$



*Example with  $b = 16$*

$$P(2 \leq X \leq 7) = \frac{5}{15} = \frac{1}{3}$$



## 11 (5) Uniform Distribution II (C2)

Let  $X$  be the delivery time (in days) for a certain shipping route. Assume

$$X \sim \text{Uniform}(a, b),$$

where  $a < 3 < 6 < b$ . Here,  $a$  is the earliest possible delivery time and  $b$  is the latest possible delivery time.

**Questions/Tasks.**

a) (2) Find the density value  $f(x)$  on the support,

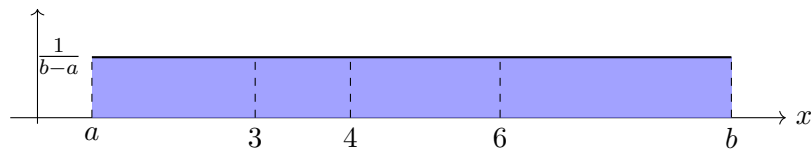
b) (2) Find  $P(3 \leq X \leq 6)$ .

c) (1) Find  $P(3 \leq X \leq 6)$  when  $a = 1, b = 9$ ,  $a = 1, b = 16$ ,  $a = 1, b = 10$  and  $a = -6, b = 9$

a) (2) Find the density value  $f(x)$  on the support,

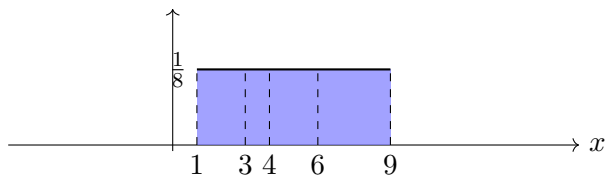
For  $X \sim \text{Uniform}(a, b)$ ,

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

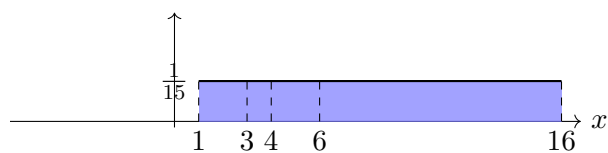


Consider the following cases obtained by assigning specific values to the endpoints.

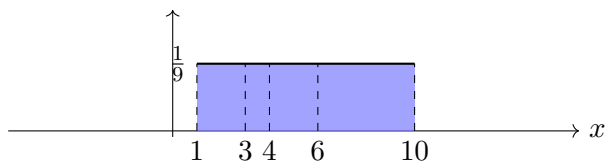
Startpoint  $a = 1$ , endpoint  $b = 9$



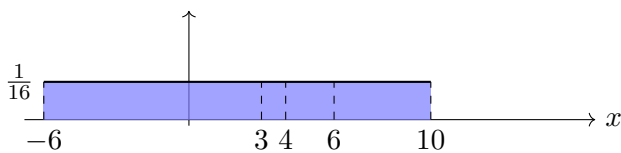
Startpoint  $a = 1$ , endpoint  $b = 16$



Endpoint  $b = 10$ , startpoint  $a = 1$

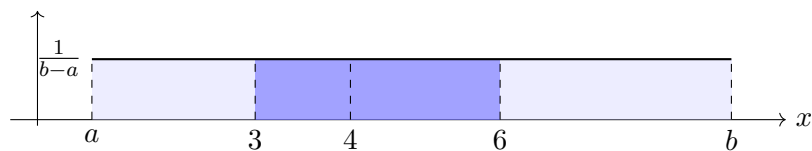


Endpoint  $b = 10$ , startpoint  $a = -6$



b) (2) Find  $P(3 \leq X \leq 6)$ .

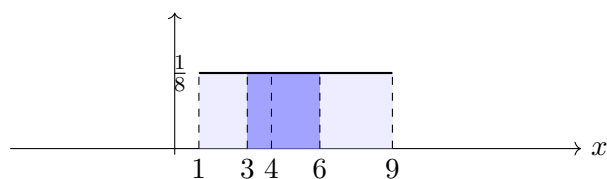
$$P(3 \leq X \leq 6) = \frac{6-3}{b-a} = \frac{3}{b-a}.$$



c) (1) Find  $P(3 \leq X \leq 6)$  when  $a = 1, b = 9$ ,  $a = 1, b = 16$ ,  $a = 1, b = 10$  and  $a = -6, b = 9$

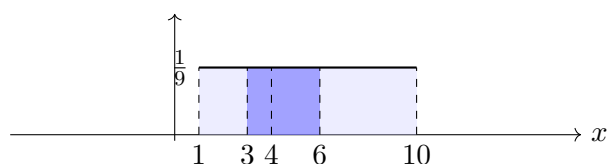
Startpoint  $a = 1$ , endpoint  $b = 9$

$$P(3 \leq X \leq 6) = \frac{3}{8}$$



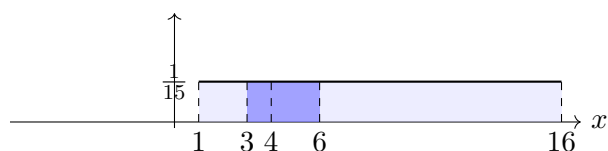
Endpoint  $b = 10$ , startpoint  $a = 1$

$$P(3 \leq X \leq 6) = \frac{3}{9} = \frac{1}{3}$$



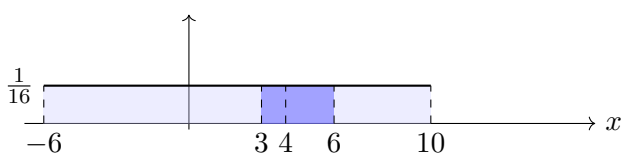
Startpoint  $a = 1$ , endpoint  $b = 16$

$$P(3 \leq X \leq 6) = \frac{3}{15} = \frac{1}{5}$$



Endpoint  $b = 10$ , startpoint  $a = -6$

$$P(3 \leq X \leq 6) = \frac{3}{16}$$



## 12 (4) Two-Interval Piecewise Uniform (C2)

**Problem.** Let  $X$  be the final value of a small short-term investment, measured in million pesos. The investor estimates that lower outcomes between 0 and 4 are more likely, while higher outcomes between 4 and 10 are less likely. The final value is modeled by the density

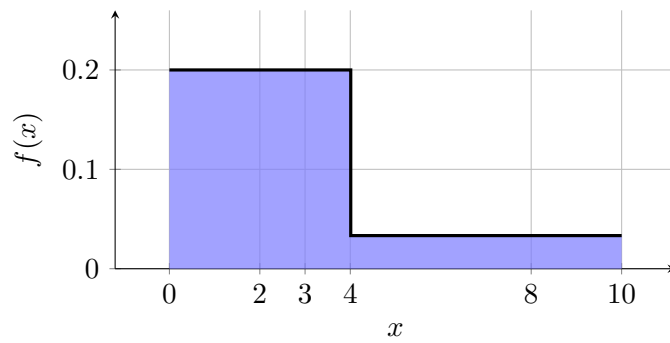
$$f(x) = \begin{cases} \frac{1}{5}, & 0 \leq x < 4, \\ \frac{1}{30}, & 4 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- a) (1) Find  $P(X < 3)$ ,
- b) (1) Find  $P(X \geq 3)$ ,
- c) (2) Find  $P(2 \leq X \leq 8)$ .

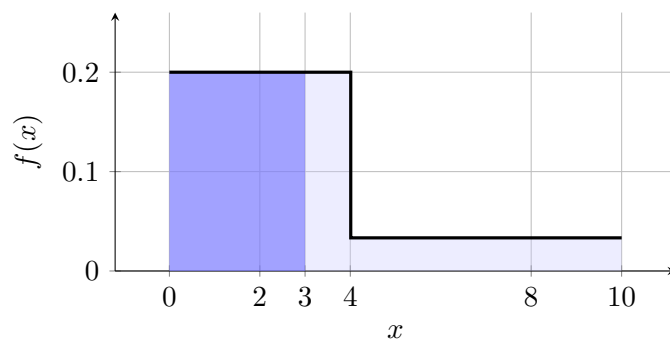
**Function definition.** Check total area:

$$4 \left( \frac{1}{5} \right) + 6 \left( \frac{1}{30} \right) = \frac{4}{5} + \frac{1}{5} = 1.$$



a) (1) Find  $P(X < 3)$ ,

$$P(X < 3) = 3 \left( \frac{1}{5} \right) = \frac{3}{5}.$$



b) (1) Find  $P(X \geq 3)$ ,

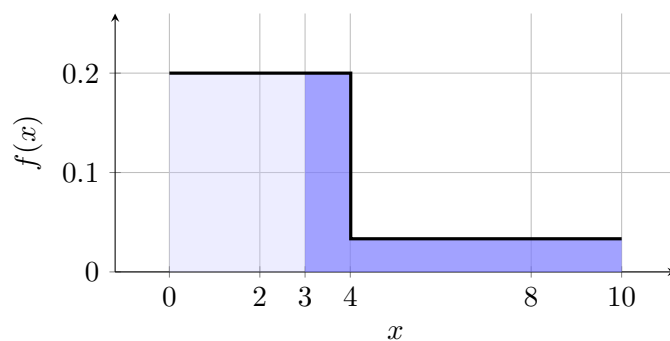
Using the standard approach,

$$P(X \geq 3) = 1 \left( \frac{1}{5} \right) + 6 \left( \frac{1}{30} \right) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}.$$

Using the complement approach,

$$P(X \geq 3) = 1 - P(X < 3) = 1 - \frac{3}{5} = \frac{2}{5}.$$

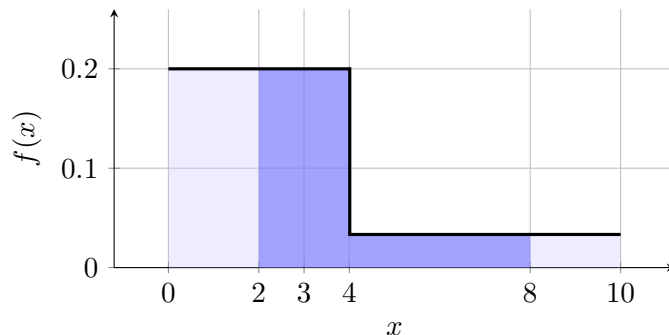
So both approaches give the same value.





c) (2) Find  $P(2 \leq X \leq 8)$ .

$$P(2 \leq X \leq 8) = 2 \left( \frac{1}{5} \right) + 4 \left( \frac{1}{30} \right) = \frac{2}{5} + \frac{2}{15} = \frac{8}{15}.$$



### 13 (5) Two-Interval Piecewise Uniform with Unique Variable Endpoint (C2)

**Problem.** Let  $X$  have density

$$f(x) = \begin{cases} \frac{1}{12}, & -2 \leq x < 1, \\ \frac{1}{8}, & 1 \leq x \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

where  $b > 1$ .

**Questions/Tasks.** Find:

- a) (2) Find the value of  $b$  so this is a valid probability distribution function,
- b) (1) Find  $P(X < -1)$ ,
- c) (1) Find  $P(X > 3)$ ,
- d) (1) Find  $P(-1 \leq X \leq 3)$ .

a) (2) Find the value of  $b$  so this is a valid probability distribution function,

For a valid density, the total area must be 1:

$$3 \left( \frac{1}{12} \right) + (b - 1) \left( \frac{1}{8} \right) = 1.$$

So

$$\frac{1}{4} + \frac{b - 1}{8} = 1.$$

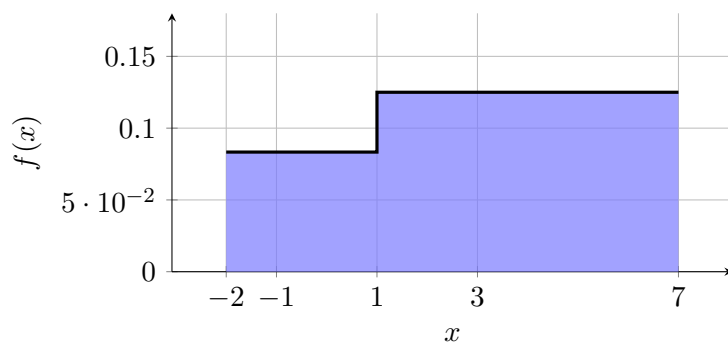
Thus

$$\frac{b - 1}{8} = \frac{3}{4},$$

and therefore

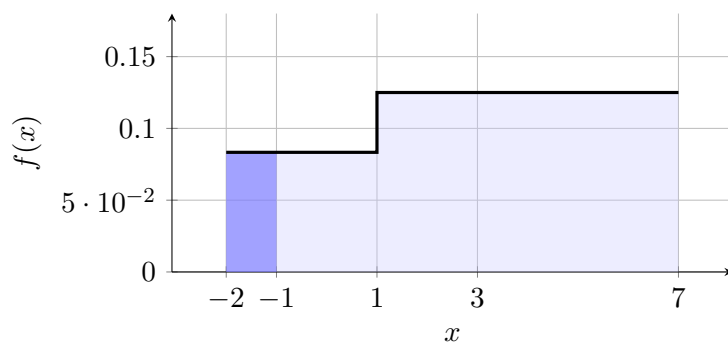
$$b - 1 = 6, \quad b = 7.$$

Now use  $b = 7$ .



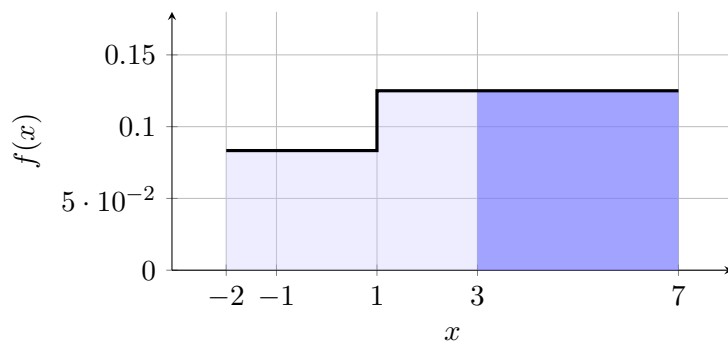
**b) (1) Find  $P(X < -1)$ ,**

$$P(X < -1) = 1 \left( \frac{1}{12} \right) = \frac{1}{12}.$$



**c) (1) Find  $P(X > 3)$ ,**

$$P(X > 3) = 4 \left( \frac{1}{8} \right) = \frac{1}{2}.$$





d) (1) Find  $P(-1 \leq X \leq 3)$ .

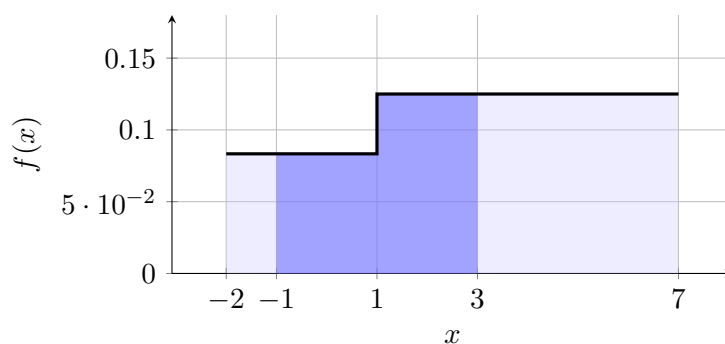
Using the standard approach,

$$P(-1 \leq X \leq 3) = 2 \left( \frac{1}{12} \right) + 2 \left( \frac{1}{8} \right) = \frac{1}{6} + \frac{1}{4} = \frac{5}{12}.$$

Using the complement approach,

$$P(-1 \leq X \leq 3) = 1 - P(X < -1) - P(X > 3) = 1 - \frac{1}{12} - \frac{1}{2} = \frac{5}{12}.$$

So both approaches give the same value.



## 14 (4) Piecewise Uniform with Unknown $k$ (C2)

**Problem.** Let  $X$  have density

$$f(x) = \begin{cases} 2k, & 0 \leq x < 2, \\ k, & 2 \leq x \leq 6, \\ 0, & \text{otherwise,} \end{cases}$$

where  $k > 0$ .

**Questions/Tasks.** Find:

- a) (2) the value of  $k$  so that  $f$  is a valid PDF,
- b) (1)  $P(X < 4)$ ,
- c) (1)  $P(4 \leq X \leq 6)$ .

a) (2) the value of  $k$  so that  $f$  is a valid PDF,

For validity, total area must be 1:

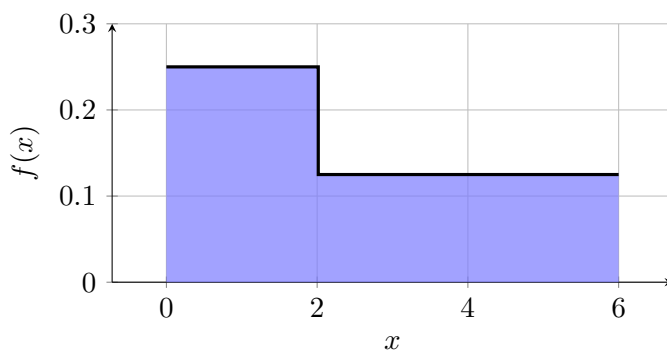
$$2(2k) + 4(k) = 1.$$

So

$$4k + 4k = 1 \quad \Rightarrow \quad 8k = 1 \quad \Rightarrow \quad k = \frac{1}{8}.$$

Therefore, the density becomes

$$f(x) = \begin{cases} \frac{1}{4}, & 0 \leq x < 2, \\ \frac{1}{8}, & 2 \leq x \leq 6, \\ 0, & \text{otherwise.} \end{cases}$$



b) (1)  $P(X < 4)$ ,

To find  $P(X < 4)$ , it is easier to use the complement.

First compute the probability of the opposite event,  $X \geq 4$ . On the interval from 4 to 6, the density is constant at  $\frac{1}{8}$ .

The width is

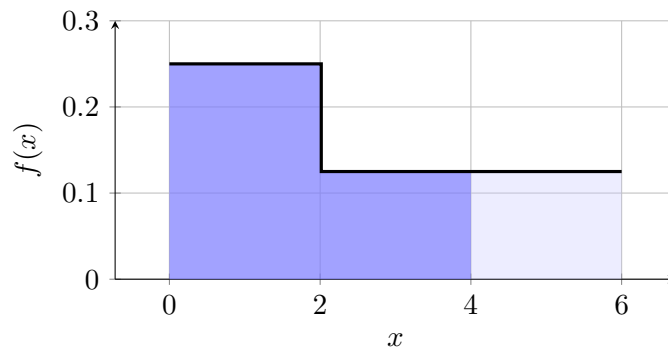
$$6 - 4 = 2,$$

so

$$P(X \geq 4) = 2 \left( \frac{1}{8} \right) = \frac{1}{4}.$$

Therefore,

$$P(X < 4) = 1 - P(X \geq 4) = 1 - \frac{1}{4} = \frac{3}{4}.$$



c) (1)  $P(4 \leq X \leq 6)$ .

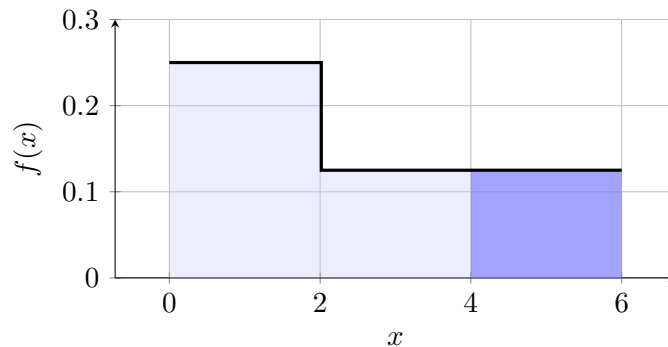
On the interval from 4 to 6, the density is constant at  $\frac{1}{8}$ .

The width is

$$6 - 4 = 2,$$

so

$$P(4 \leq X \leq 6) = 2 \left( \frac{1}{8} \right) = \frac{1}{4}.$$



## 15 (12) Small-Business Investment(C3)

An investor invests in a short-term small-business opportunity: buying low-cost clothing bundles to resell online and at weekend markets. They estimate the final value cannot be below zero million pesos because, in this simplified model, the end-of-period value cannot be negative. They also estimate it cannot exceed thirty million pesos because their first-round sales capacity is limited. Their most likely final value is twelve million pesos, based on expected demand from regular customers. Values below twelve million pesos are possible, but become less likely as results move farther down from that level. Values above twelve million pesos are also possible, but after that point, very high outcomes become gradually less likely. Since the model is continuous, the probability of obtaining any exact extreme value can be considered 0, so the probabilities of ending at exactly zero million pesos or exactly thirty million pesos are treated

as 0. The initial investment is six million pesos. A loss occurs if the final value is below that amount, and an earning occurs if the final value is above it.

**Questions/Tasks.**

- a) (4) Write the pdf as a piecewise function.
- b) (1) Verify that the total area under the density is one.
- c) (3) Find the probability of loss and the probability of earning.
- d) (4) Find the specific probability of earning more than 50% of the investment.

**a) (4) Write the pdf as a piecewise function.**

Let  $V$  be the final value of the investment, measured in million pesos.

The support is from 0 to 30, and the most likely value is 12. Since the model is triangular, the graph rises linearly from  $(0, 0)$  to the peak at  $(12, h)$ , then falls linearly from  $(12, h)$  to  $(30, 0)$ .

First determine the peak height  $h$  from total area = 1. The whole graph is one triangle with base

$$30 - 0 = 30.$$

So

$$\frac{1}{2}(30)(h) = 1.$$

Hence

$$15h = 1 \quad \Rightarrow \quad h = \frac{1}{15}.$$

Now write each linear piece.

On the interval from 0 to 12, the line goes through  $(0, 0)$  and  $(12, \frac{1}{15})$ . Its slope is

$$\frac{\frac{1}{15} - 0}{12 - 0} = \frac{1}{180}.$$

So the left-hand piece is

$$f(v) = \frac{v}{180} \quad \text{for } 0 \leq v \leq 12.$$

On the interval from 12 to 30, the line goes through  $(12, \frac{1}{15})$  and  $(30, 0)$ . Its slope is

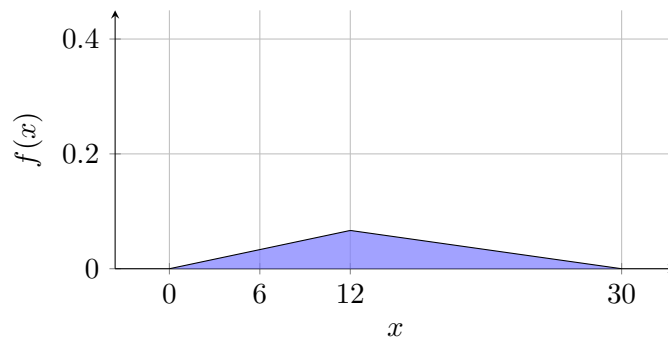
$$\frac{0 - \frac{1}{15}}{30 - 12} = -\frac{1}{270}.$$

So the right-hand piece is

$$f(v) = \frac{30 - v}{270} \quad \text{for } 12 \leq v \leq 30.$$

Therefore, the pdf is

$$f(v) = \begin{cases} \frac{v}{180}, & 0 \leq v \leq 12, \\ \frac{30 - v}{270}, & 12 \leq v \leq 30, \\ 0, & \text{otherwise.} \end{cases}$$



**b) (1) Verify that the total area under the density is one.**

The graph is a single triangle with base 30 and height  $\frac{1}{15}$ . Therefore,

$$\text{Total area} = \frac{1}{2}(30) \left( \frac{1}{15} \right) = 1.$$

So this is a valid pdf.



**c) (3) Find the probability of loss and the probability of earning.**

**Probability of loss.** A loss occurs when the final value is below the purchase price of 6 million pesos. On the graph, this is the small triangle from 0 to 6.

Its base is

$$6 - 0 = 6.$$

Its height is

$$f(6) = \frac{6}{180} = \frac{1}{30}.$$

So

$$P(\text{loss}) = P(V < 6) = \frac{1}{2}(6) \left( \frac{1}{30} \right) = \frac{6}{60} = \frac{1}{10}.$$

**Probability of earning.** An earning occurs when the final value is above the purchase price of 6 million pesos. Since the total probability is 1 and a loss occurs when the final value is below 6, we use the complement rule

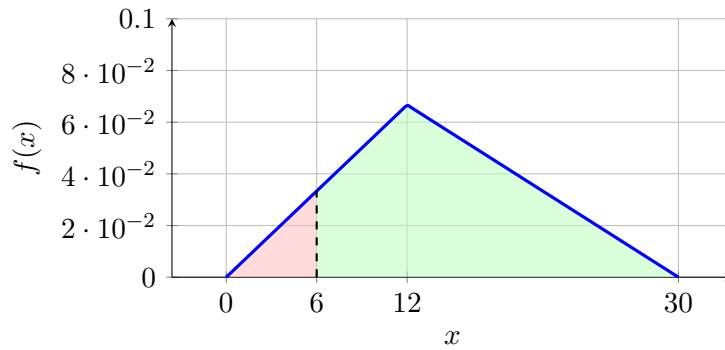
$$P(\text{earning}) = P(V > 6) = 1 - P(V < 6).$$

From the previous result,

$$P(V < 6) = \frac{1}{10}.$$

Therefore,

$$P(V > 6) = 1 - \frac{1}{10} = \frac{9}{10}.$$



d) (4) Find the specific probability of earning more than 50% of the investment.

The investment amount is 6 million pesos, so earning more than 50% means ending above

$$6(1.5) = 9.$$

Therefore, we need

$$P(\text{earning more than 50\%}) = P(V > 9).$$

Notice that 9 lies to the left of the most probable value 12. For that reason, the complement rule is convenient here, because the excluded region from 0 to 9 is a simple triangle.

So

$$P(V > 9) = 1 - P(V \leq 9).$$

On the interval from 0 to 9, the graph forms a triangle.

Its base is

$$9 - 0 = 9,$$

and its height is

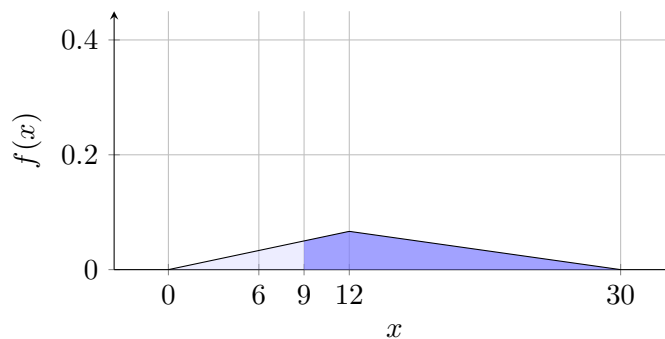
$$f(9) = \frac{9}{180} = \frac{1}{20}.$$

Hence

$$P(V \leq 9) = \frac{1}{2}(9) \left( \frac{1}{20} \right) = \frac{9}{40}.$$

Therefore,

$$P(V > 9) = 1 - \frac{9}{40} = \frac{31}{40}.$$







## 16 (11) Basic Symmetric Triangular Density (C3)

A random variable  $Y$  is modeled on the support  $[0, 8]$  by one of the following two functions:

$$f(y) = \begin{cases} \frac{y}{16}, & 0 \leq y \leq 4, \\ \frac{8-y}{16}, & 4 \leq y \leq 8, \\ 0, & \text{otherwise,} \end{cases} \quad g(y) = \begin{cases} \frac{y}{12}, & 0 \leq y \leq 4, \\ \frac{8-y}{12}, & 4 \leq y \leq 8, \\ 0, & \text{otherwise.} \end{cases}$$

### Questions/Tasks.

- (a) **(3)** Decide which of the two functions is a valid probability density function on the interval  $[0, 8]$ . Justify your answer.
- (b) **(4)** For the valid density, compute  $P(Y > 2)$ .
- (c) **(4)** For the valid density, find the most probable interval of length 2, compute its probability, and draw the corresponding graph.
- a) (3) Decide which of the two functions is a valid probability density function on the interval  $[0, 8]$ . Justify your answer.**

Both functions are nonnegative on  $[0, 8]$ , so we only need to check total area.

For  $f(y)$ , the graph is a triangle with base 8 and peak height

$$f(4) = \frac{4}{16} = \frac{1}{4}.$$

So

$$\text{Area under } f = \frac{1}{2}(8) \left( \frac{1}{4} \right) = 1.$$

Therefore,  $f(y)$  is a valid PDF.

For  $g(y)$ , the graph is a triangle with base 8 and peak height

$$g(4) = \frac{4}{12} = \frac{1}{3}.$$

So

$$\text{Area under } g = \frac{1}{2}(8) \left( \frac{1}{3} \right) = \frac{4}{3} \neq 1.$$

Therefore,  $g(y)$  is not a valid PDF.



**b) (4) For the valid density, compute  $P(Y > 2)$ .**

Use the complement with the valid density  $f(y)$ :

$$P(Y > 2) = 1 - P(0 \leq Y \leq 2).$$

The area from 0 to 2 is a triangle with base 2 and height

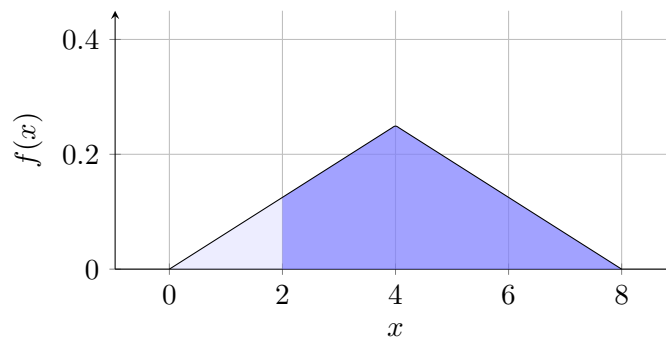
$$f(2) = \frac{2}{16} = \frac{1}{8}.$$

So

$$P(0 \leq Y \leq 2) = \frac{1}{2}(2) \left( \frac{1}{8} \right) = \frac{1}{8}.$$

Hence

$$P(Y > 2) = 1 - \frac{1}{8} = \frac{7}{8}.$$



**c) (4) For the valid density, find the most probable interval of length 2, compute its probability, and draw the corresponding graph.**

For the valid density  $f(y)$ , the most probable interval of length 2 is the one centered at the peak, namely

$$[3, 5].$$

This is because the density increases up to  $y = 4$  and then decreases after  $y = 4$ , so an interval of fixed length captures the most area when it is placed around the highest point of the graph.

Now compute its probability using the complement rule. The interval

$$[3, 5]$$

leaves two end regions:

$$[0, 3] \quad \text{and} \quad [5, 8].$$

So

$$P(3 \leq Y \leq 5) = 1 - P(0 \leq Y \leq 3) - P(5 \leq Y \leq 8).$$

The region from 0 to 3 is a triangle with base

and height

$$f(3) = \frac{3}{16}.$$

So

$$P(0 \leq Y \leq 3) = \frac{1}{2}(3) \left( \frac{3}{16} \right) = \frac{9}{32}.$$

The region from 5 to 8 is also a triangle with base

$$8 - 5 = 3$$

and height

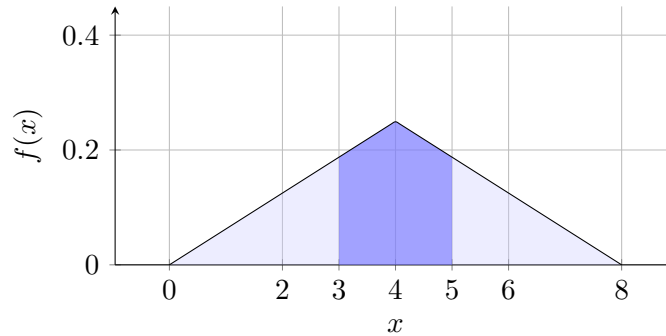
$$f(5) = \frac{8-5}{16} = \frac{3}{16}.$$

So

$$P(5 \leq Y \leq 8) = \frac{1}{2}(3) \left( \frac{3}{16} \right) = \frac{9}{32}.$$

Therefore,

$$P(3 \leq Y \leq 5) = 1 - \frac{9}{32} - \frac{9}{32} = 1 - \frac{18}{32} = \frac{14}{32} = \frac{7}{16}.$$



## 17 (12) Introductory Linear Density (C3)

Let  $X$  be a continuous random variable with density

$$f(x) = kx, \quad 0 \leq x \leq b,$$

and  $f(x) = 0$  otherwise, where  $b > 0$  and  $k > 0$ .

**Questions/Tasks.**

- (a) **(3)** Find  $k$  geometrically so that  $f$  is a valid PDF, and draw the corresponding graph.
- (b) **(3)** Compute  $P(0 \leq X \leq a)$ , where  $0 \leq a \leq b$ , and draw the corresponding graph.
- (c) **(3)** Compute  $P(a \leq X \leq b)$ , where  $0 \leq a \leq b$ , and draw the corresponding graph.
- (d) **(3)** Compute  $P(a \leq X \leq c)$ , where  $0 \leq a \leq c \leq b$ , and draw the corresponding graph.

a) (3) Find  $k$  geometrically so that  $f$  is a valid PDF, and draw the corresponding graph.

The graph is a single straight line from  $(0,0)$  to  $(b, kb)$ . So total area is triangle area:

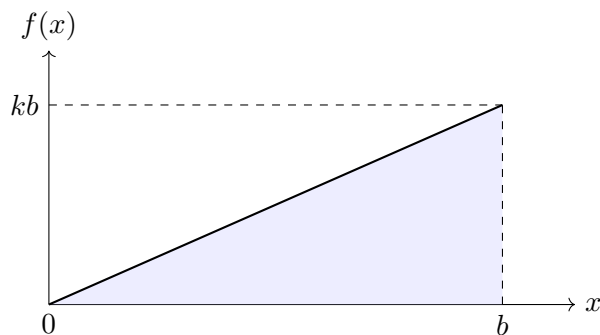
$$\frac{1}{2}(b)(kb) = 1.$$

Hence

$$\frac{kb^2}{2} = 1 \quad \Rightarrow \quad k = \frac{2}{b^2}.$$

Therefore,

$$f(x) = \frac{2x}{b^2}, \quad 0 \leq x \leq b.$$



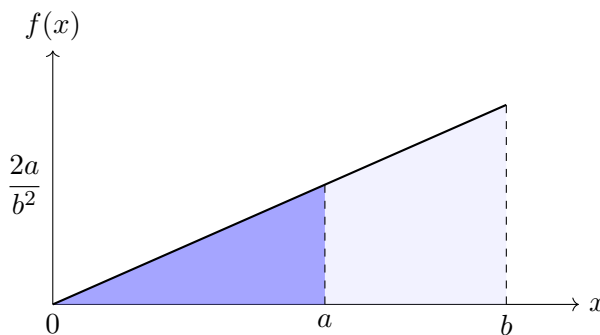
b) (3) Compute  $P(0 \leq X \leq a)$ , where  $0 \leq a \leq b$ , and draw the corresponding graph.

On  $[0, a]$ , the area is a smaller triangle with height

$$f(a) = \frac{2a}{b^2}.$$

So

$$P(0 \leq X \leq a) = \frac{1}{2}(a) \left( \frac{2a}{b^2} \right) = \frac{a^2}{b^2}.$$



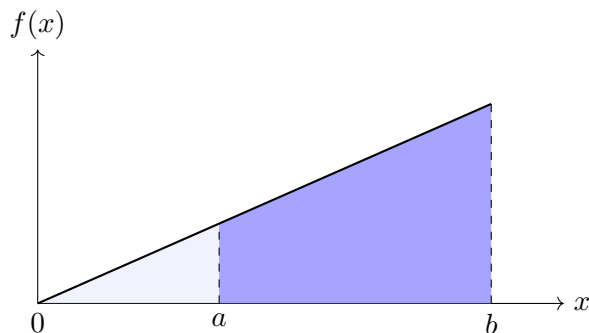
c) (3) Compute  $P(a \leq X \leq b)$ , where  $0 \leq a \leq b$ , and draw the corresponding graph.

To find  $P(a \leq X \leq b)$ , subtract the area from 0 to  $a$  from the total area:

$$P(a \leq X \leq b) = 1 - P(0 \leq X \leq a).$$

Using part (b),

$$P(a \leq X \leq b) = 1 - \frac{a^2}{b^2} = \frac{b^2 - a^2}{b^2}.$$



d) (3) Compute  $P(a \leq X \leq c)$ , where  $0 \leq a \leq c \leq b$ , and draw the corresponding graph.

To find  $P(a \leq X \leq c)$ , subtract the area from 0 to  $a$  from the area from 0 to  $c$ :

$$P(a \leq X \leq c) = P(0 \leq X \leq c) - P(0 \leq X \leq a).$$

Using part (b),

$$P(0 \leq X \leq c) = \frac{c^2}{b^2} \quad \text{and} \quad P(0 \leq X \leq a) = \frac{a^2}{b^2}.$$

Therefore,

$$P(a \leq X \leq c) = \frac{c^2}{b^2} - \frac{a^2}{b^2} = \frac{c^2 - a^2}{b^2}.$$

